Analyzing Speech for Pronunciation Tasks

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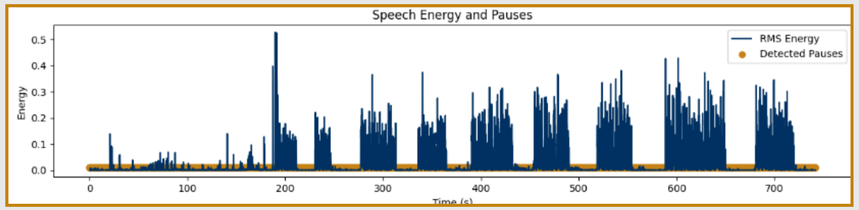
Background

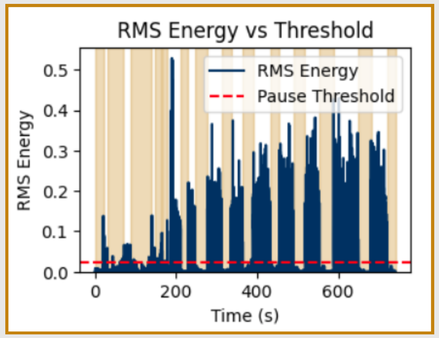
- In many interview processes for tech roles, candidates must send in video responses.
- Using English pronunciation assessments, companies can screen candidates and filter out/flag suspicious speech behavior or candidates who are not proficient in English.
- By filtering and flagging candidates automatically, recruiters get to save time and optimize the interview process.

First Steps

- Started with a basic speech energy model using RMS energy to detect pauses in audio recordings.
- Low-energy segments were identified as pauses, providing insight into the speaker's rhythm and fluency.



Identifying Pauses



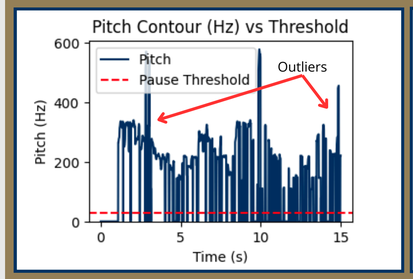
- Most pauses were detected using a threshold set at 5% of the max RMS energy.
- Continuous low-energy regions were marked as candidate pauses.
- Pauses shorter than 10 seconds were filtered out.
- In the plot, shaded regions show where silence occurred between spoken segments.

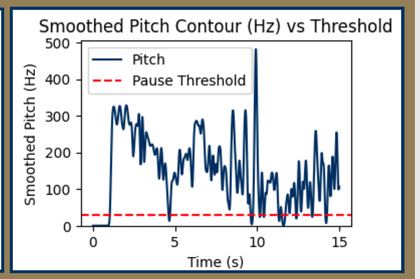
Identifying and Splitting Speech Segments

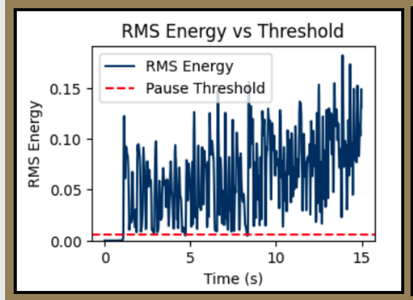
- Any audio in between pauses was considered a speech segment.
- Speech segments were isolated for individual analysis
- Segments were further split into 2-5 second audio clips to get pitch range and syllable rate metrics for each clip.

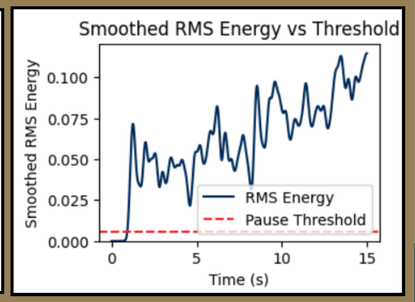
Smoothing to Remove Outliers

- The analysis of pitch contour was skewed by outliers, making it harder to find patterns in the audio.
- A Gaussian smoothing function was implemented.
- Different sigma values were evaluated. Sigma = 5 was selected based on the smoothing and retention of key information.
- Top Plot: Smoothing pitch contour to remove outliers using sigma = 5.
- Bottom Plot: Smoothing RMS Energy to remove outliers using sigma = 10.

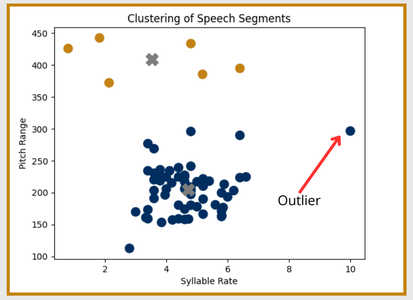


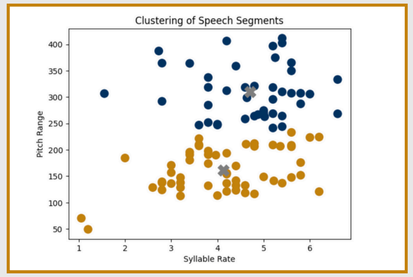






K-Means Clustering





- Different values of k were evaluated based on inertia and visual assessment of clustering performance.
- In the first plot, a majority of points had pitch ranges below 300, with a few outlying points in the 350-450 range.
- There is one data point with an unusually high syllable rate value.
- In the second audio recording, clustering did not reveal outliers, suggesting that there was no suspicious speech.

Takeaways

- Smoothing pitch and energy curves reduced noise and revealed clearer speech patterns.
- Pause tuning prevented over-segmentation, preserving natural speech flow.
- Simple features like average pitch and syllable rate effectively summarized speaking style.
- K-means clustering distinguished typical vs. atypical speech segments.
- Small tuning choices greatly impacted clustering, showing how critical they are for capturing meaningful speech patterns.
- Future studies should examine:
  - The causes of outliers in pitch contour.
  - Additional features that are useful in flagging suspicious speech.
  - Other methods for clustering and identifying outliers.

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